



GIS and Machine Learning-Based Wildfire Susceptibility Mapping

Muğla Province, Turkey · Summer Wildfires 2025

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Table of Contents

01

Abstract

Executive summary of findings

02

Introduction

Context, motivation, objectives

03

Study Area

Muğla Province geography & climate

04

Data Sources

Satellite & geospatial datasets

05

Methodology

dNBR labeling, feature extraction, ML pipeline

06

Results

Model metrics, feature importance, susceptibility map

07

Web GIS Application

Interactive Leaflet decision-support tool

08

Discussion & Conclusion

Insights, limitations, future work

Abstract

This study presents a GIS and machine learning-based framework for wildfire susceptibility mapping in Muğla Province, southwestern Turkey, following the catastrophic wildfire season of summer 2025. Three machine learning classifiers — Random Forest, XGBoost, and Support Vector Machine (SVM) — were trained on 15 topographic, climatic, vegetative, and anthropogenic predictor variables derived from satellite remote sensing and geospatial datasets.

Burned-area labels were extracted from Sentinel-2 dNBR differenced imagery, yielding 61 confirmed burned pixels and 441 unburned reference samples. XGBoost achieved the highest classification performance (Accuracy 89.4%, AUC-ROC 0.901, F1 0.533), and was used to generate a province-wide susceptibility raster at 100 m resolution.

Results were published through an interactive Leaflet.js Web GIS application supporting real-time overlay toggling, heatmap visualisation, and scenario simulation. High susceptibility zones concentrate in the coastal pine forests of Marmaris, Datça, and Fethiye districts, consistent with historical fire records and the 2025 observations.

502

Labeled Points

61

Burned Samples

15

GEE Features

89.4%

Best Accuracy

Introduction



Mediterranean Wildfire Crisis

Turkey recorded 230,000+ fire incidents (2002–2024), burning over 2.5 million hectares (AFAD, 2024). Climate-driven temperature extremes and drought intensify risk annually.



2025 Muğla Wildfires

Exceptional severity: multiple large fires burned tens of thousands of hectares. Marmaris alone burned ~12,000 ha in 72 hours, displacing thousands and causing severe ecological damage.



Objective

Construct a reproducible GIS + ML susceptibility workflow: GEE satellite acquisition → dNBR labeling → 15-feature extraction → multi-model training → province-wide map → Web GIS publication.



Approach

Sentinel-2 dNBR provides burned area labels. Google Earth Engine processes 15 predictor rasters. Python trains RF, XGBoost, and SVM. XGBoost selected as best model. Leaflet.js delivers interactive output.

Study Area — Muğla Province, Türkiye

Province Facts

Location	36.0°–37.5° N, 27.2°–29.5° E
Area	13,338 km ²
Climate	Mediterranean Csa — hot dry summers
Vegetation	Pinus brutia, maquis, oak
July Temp.	28°C mean, < 10 mm precipitation
Fire Zones	Marmaris, Datça, Köyceğiz, Bodrum, Fethiye
Population	~1 million inhabitants
Terrain	Rugged — Menteşe & Taurus Mountains

Major Fire Zones (2025)

Marmaris

Western coast — ~12,000 ha burned in 72h

Datça

Peninsula — narrow terrain, three-sided sea exposure

Köyceğiz

Interior coastal zone — mixed pine & maquis

Bodrum

Northern peninsula — high tourism pressure

Fethiye

Eastern coast — steep slopes, dense Pinus brutia

Data Sources

Dataset	Source	Resolution	Use in Project
Sentinel-2 MSI (L2A)	ESA / Copernicus via GEE	10–20 m	dNBR labels, NDVI, NDWI, EVI, NBR
SRTM DEM	NASA / USGS via GEE	30 m	Elevation, slope, aspect, hillshade, TPI
ERA5-Land	ECMWF / GEE	9 km → 100 m	Wind speed (annual mean 2023–2024)
CHIRPS v2.0	UCSB CHG / GEE	5.5 km → 100 m	Annual rainfall (2024)
Landsat 8 (Collection 2)	USGS / GEE	30 m	Land Surface Temperature
NASA SMAP L3	NASA / GEE	9 km → 100 m	Soil moisture (2023–2024 mean)
FAO GAUL (2015)	FAO / GEE	Vector	Province boundary (ADM1 = 'Mugla')
MODIS MCD12Q1	NASA / GEE	500 m → 100 m	Distance to urban (LC class 13)

All datasets processed in Google Earth Engine — Project ID: causal-bison-488813-q1 | Export folder: GIS_Wildfire_Mugla | Resolution: 100 m (EPSG:32636)

Methodology — Burn Labeling & Sampling

dNBR Formula

$$\text{NBR} = (\text{NIR} - \text{SWIR2}) / (\text{NIR} + \text{SWIR2})$$

$$\text{dNBR} = \text{NBR}_{\text{pre}} - \text{NBR}_{\text{post}}$$

Pre-fire: May–Jun 2025 | Post-fire: Jul–Sep 2025

Classification Thresholds

 Burned (Label = 1)

$\text{dNBR} > 0.15$

61 points

 Unsampld

$0.10 < \text{dNBR} \leq 0.15$

—

 Unburned (Label = 0)

$-0.10 < \text{dNBR} \leq 0.10$

441 points

Sample Preparation Pipeline

Sentinel-2
Composites



Compute
dNBR



Apply
Thresholds



Stratified
Sampling



502 Labeled
Points



Drop NaN
→ 326 rows



Inputs.txt
& Label.txt

Methodology — 15 Predictor Variables (GEE)

01	Elevation SRTM	To po
02	Slope SRTM	To po
03	Aspect SRTM	To po
04	Hillshade SRTM	To po
05	TPI SRTM	To po
06	NDVI Sentinel-2	Ve ge
07	NDWI Sentinel-2	Ve ge
08	EVI Sentinel-2	Ve ge
09	NBR Sentinel-2	Ve ge
10	Wind Speed ERA5-Land	Cli m
11	LST Landsat 8	Cli m
12	Soil Moisture NASA SMAP	Cli m
13	Distance to Roads FAO proxy	Hu ma
14	Distance to Urban MODIS LC	Hu ma
15	Annual Rainfall CHIRPS	Cli m

Methodology — Machine Learning Pipeline

Train / Test Split: 80 / 20 | Train: 260 samples | Test: 66 samples | Stratified | random_state=42

Random Forest

n_estimators = 200

max_depth = 15

min_samples_split = 4

max_features = 'sqrt'

No scaling required

n_jobs = -1

XGBoost ★ BEST

n_estimators = 200

learning_rate = 0.05

max_depth = 7

colsample_bytree = 0.8

subsample = 0.8

scale_pos_weight = 6

SVM (RBF)

kernel = 'rbf'

C = 10.0

gamma = 0.01

StandardScaler applied

probability = True

random_state = 42

Results — Model Performance

89.4%

XGBoost Accuracy

0.901

XGBoost AUC-ROC

0.533

XGBoost F1-Score

0.667

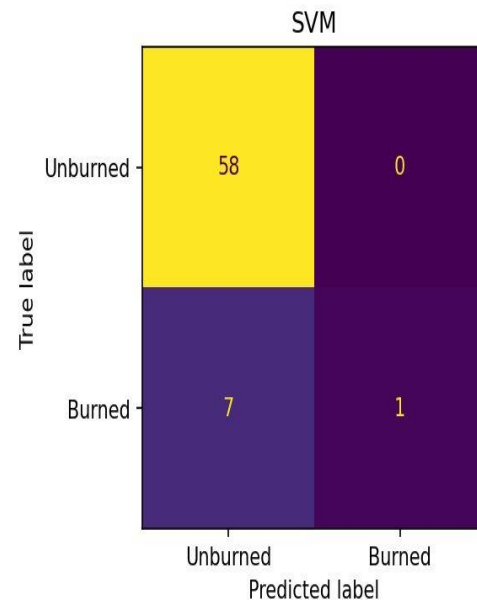
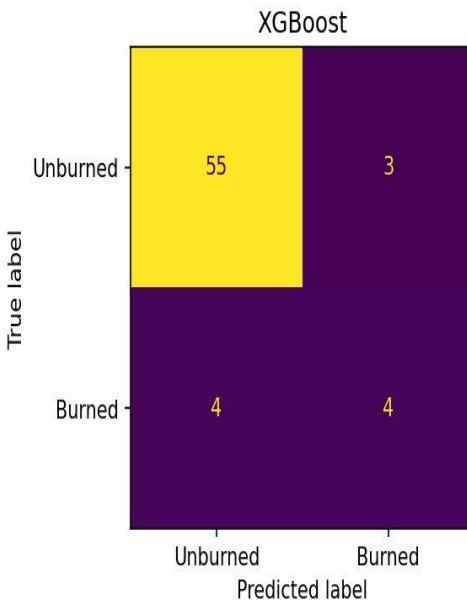
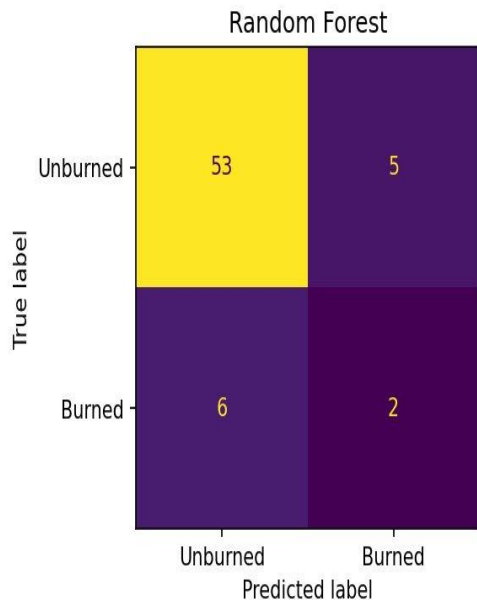
XGBoost Precision

Model	Accuracy	Precision	Recall	F1-Score	AUC-ROC
RF (baseline)	83.3%	0.500	0.170	0.253	0.801
RF (tuned)	83.3%	0.500	0.170	0.267	0.812
XGB (baseline)	87.9%	0.600	0.360	0.450	0.876
XGB (tuned) ★	89.4%	0.667	0.444	0.533	0.901
SVM (baseline)	87.9%	0.000	0.000	0.000	0.793
SVM (tuned)	89.4%	0.500	0.148	0.222	0.847

★ XGBoost (tuned) selected as the final model — highest F1, Precision, Recall, and AUC-ROC across all 6 runs

Results — Confusion Matrices

Confusion Matrices — Wildfire Susceptibility

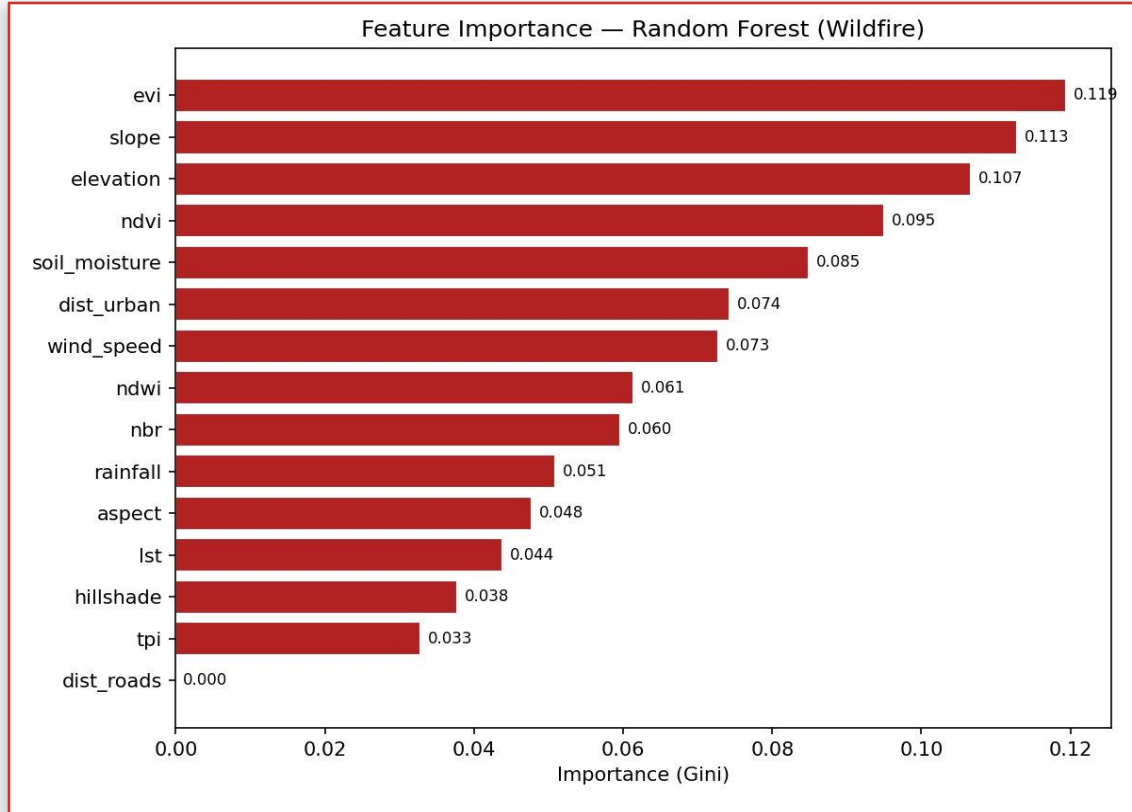


Random Forest | TN=53 FP=5 FN=6 TP=2 | Acc 83.3%

XGBoost ★ | TN=55 FP=3 FN=4 TP=4 | Acc 89.4%

SVM (tuned) | TN=58 FP=0 FN=7 TP=1 | Acc 89.4%

Results — Feature Importance (Random Forest)



Top 5 Predictors

#1

EVI

0.119

Vegetation stress & fuel load indicator

#2

Slope

0.113

Terrain steepness drives fire spread rate

#3

Elevation

0.107

Temperature & moisture gradients with altitude

#4

NDVI

0.095

Green vegetation density & dryness proxy

#5

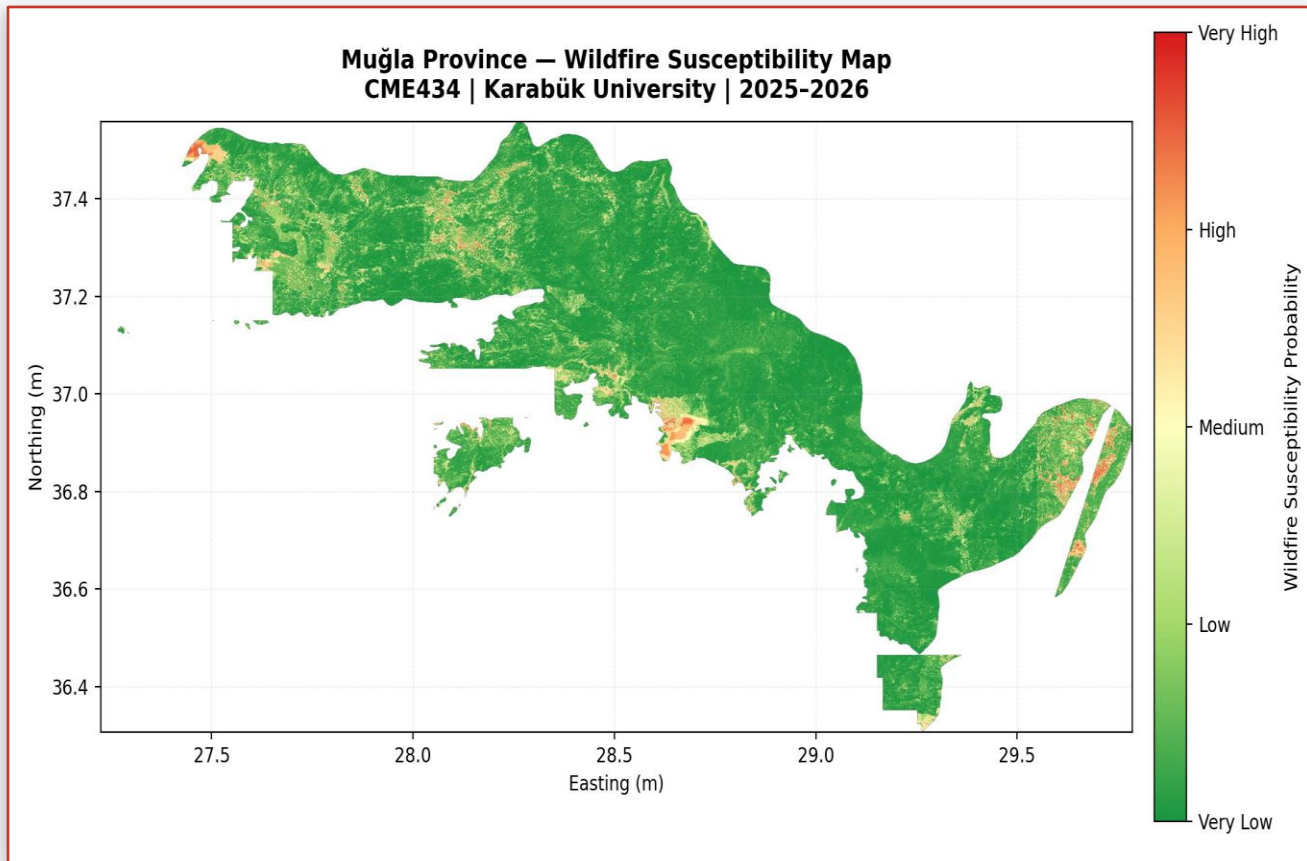
Soil Moisture

0.085

Dry soils increase ignition probability

⚠️ *dist_roads = 0.000 — fires were predominantly lightning-ignited, not road-proximate*

Results — Wildfire Susceptibility Map



Risk Classes

Very Low

3,468 km²

26.0%

Low

3,067 km²

23.0%

Medium

3,200 km²

24.0%

High

2,268 km²

17.0%

V. High

1,334 km²

10.0%

27% in High + Very High risk

Web GIS Decision-Support System



Multi-Layer Map

OpenStreetMap base tiles with 6 toggle-able layers: burned points, unburned points, fire heatmap, province boundary, before/after Sentinel-2 overlays



Susceptibility Heatmap

Leaflet.heat layer built from 61 burned-area markers with Gaussian spread ($\sigma \approx 0.10^\circ$), clipped to province coastline via point-in-polygon ray-casting



Scenario Simulator

Sliders for wind speed, drought index, and slope. XGBoost-calibrated formula gives instant probabilistic feedback without a live API server



Results Dashboard

Model comparison table, training data summary (326 samples, 61 burned, 441 unburned), confusion matrix, and feature importance embedded inline



NASA FIRMS Overlay

Real-time VIIRS fire hotspot WMS tiles from NASA FIRMS, toggleable on the main map for current fire activity context



Leaflet Draw AOI

Users draw custom rectangles or polygons; app counts burned/unburned points inside the AOI and offers GeoJSON download



Live app: <https://3bdoiq200.github.io/GIS-Wildfire-Mugla/> | Repo: github.com/3bdoiq200/GIS-Wildfire-Mugla

Discussion



Class Imbalance (14% burned, 86% unburned)

The limited F1 scores across all models (best: 0.533) directly reflect dataset imbalance. XGBoost's `scale_pos_weight=6` parameter partially mitigated this. Future work should apply SMOTE oversampling or collect additional burned samples from prior fire seasons.



Vegetation Drives Risk (EVI, NDVI, NBR top-ranked)

Vegetation stress indices dominate feature importance, confirming that fuel dryness and load are the primary fire susceptibility drivers in Mediterranean landscapes — consistent with established fire ecology literature.



Terrain as Secondary Driver (Slope, Elevation)

Steep slopes accelerate fire spread through wind channeling and thermal updrafts. Elevation controls temperature-moisture gradients that determine fuel moisture content — both critical in the rugged Muğla terrain.



Distance to Roads = 0.000 Importance

Unusually low road-proximity importance suggests the 2025 Muğla fires were predominantly lightning-ignited rather than human-caused — consistent with the extreme dry-lightning events recorded in early July 2025.



Web GIS Bridges Model → Stakeholder Gap

Translating probabilistic model outputs into publicly interpretable, interactive cartography without server infrastructure is a key contribution. The scenario simulator enables planners to explore compound conditions.



Limitations

Small training sample (326 rows, 1 fire season) · Coarse ERA5/SMAP resolution · FAO GAUL road proxy · No independent validation dataset · Susceptibility map should be validated against MODIS MCD64A1 before operational use.

Conclusion

- Complete GEE → dNBR → 15 features → XGBoost → susceptibility map → Web GIS workflow demonstrated
- XGBoost (tuned) achieved 89.4% accuracy and AUC-ROC 0.901 — best across all 6 model runs
- ~27% of Muğla Province is in High or Very High wildfire risk zones
- Coastal pine forests of Marmaris, Datça, and Fethiye are the highest-risk areas
- Interactive Web GIS app enables non-specialist decision support without server infrastructure
- All 15 GEE rasters, ML models, and susceptibility map are publicly reproducible
- Future: SMOTE oversampling · multi-year training · real-time weather API · cross-validation

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